

## Performance Testing

|               |                                  |
|---------------|----------------------------------|
| Date          | 12 August, 2026                  |
| Team ID       | Individual                       |
| Project Name  | AI Knowledge Retention Predictor |
| Maximum Marks | 5 Marks                          |

### Performance Testing

Performance Testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field               | Details                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------|
| Testing Tool Used   | Postman & Locust                                                                        |
| Type of Testing     | Load Testing & API Response Time Testing                                                |
| Target Module / API | Streamlit Web App (/) & Google Gemini API Integration ( <code>generate_content</code> ) |
| Test Environment    | Cloud (Streamlit Community Cloud) & Local Deployment                                    |
| Test Date           | 17 August 2026                                                                          |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description                                      | No. of Virtual Users | Duration (sec) | Expected Outcome                                    |
|------|------------------------------------------------------------------|----------------------|----------------|-----------------------------------------------------|
| 1.   | Concurrent user access to mathematical retention curve dashboard | 20                   | 60             | Page renders graph seamlessly without lag (< 1.5s). |

|    |                                                                |    |     |                                                                            |
|----|----------------------------------------------------------------|----|-----|----------------------------------------------------------------------------|
| 2. | Rapid quiz generation calls via Gemini API integration         | 10 | 120 | Returns 5 structured quiz questions without 401 or rate-limit errors.      |
| 3. | Database read/write operations with concurrent SQLite sessions | 15 | 60  | Quiz scores and user inputs write cleanly to SQLite without database lock. |
| 4. | Peak traffic spike on Streamlit web app UI                     | 50 | 180 | App maintains server availability with < 3% error rate.                    |

Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                                                  |
|------|----------------------|--------------|--------------|----------------------|----------------------------------------------------------|
| 1.   | Response Time (Avg)  | < 2 seconds  | 1.24 seconds | Pass                 | UI components load smoothly under initial load.          |
| 2.   | Response Time (Max)  | < 5 seconds  | 3.85 seconds | Pass                 | Occurs during cold-start API quiz generation.            |
| 3.   | Throughput (Req/sec) | > 10 req/sec | 14.2 req/sec | Pass                 | Handled concurrency effectively on Cloud.                |
| 4.   | Error Rate           | < 1%         | 0.2%         | Pass                 | Zero authentication timeouts after API key fix.          |
| 5.   | CPU Utilization      | < 80%        | 42%          | Pass                 | Minimal overhead during mathematical decay calculations. |
| 6.   | Memory Utilization   | < 80%        | 58%          | Pass                 | Dynamic memory clearing prevented leaks.                 |

Step 4: Observations & Analysis

Key Findings:

The application demonstrates stable performance under normal and moderate peak

loads. The Ebbinghaus forgetting curve computations execute near-instantly in memory. Gemini API responses average around ~2.8 seconds for multi-question generation, which aligns with expected cloud LLM latency.

**Bottlenecks Identified:**

Synchronous API calls to Google Gemini create a minor UI spinner delay during quiz generation under high concurrent requests. Local SQLite database writes can experience occasional locking if multiple user threads attempt simultaneous writes.

**Optimization Steps Taken:**

- Implemented explicit dynamic client initialization for `google-generativeai` to prevent cached unauthenticated states (`401 UNAUTHENTICATED`).
- Optimized Streamlit session state management to minimize unnecessary full-page rerenders.
- Structured light SQLite connections that open and close explicitly per operation to avoid connection locks.